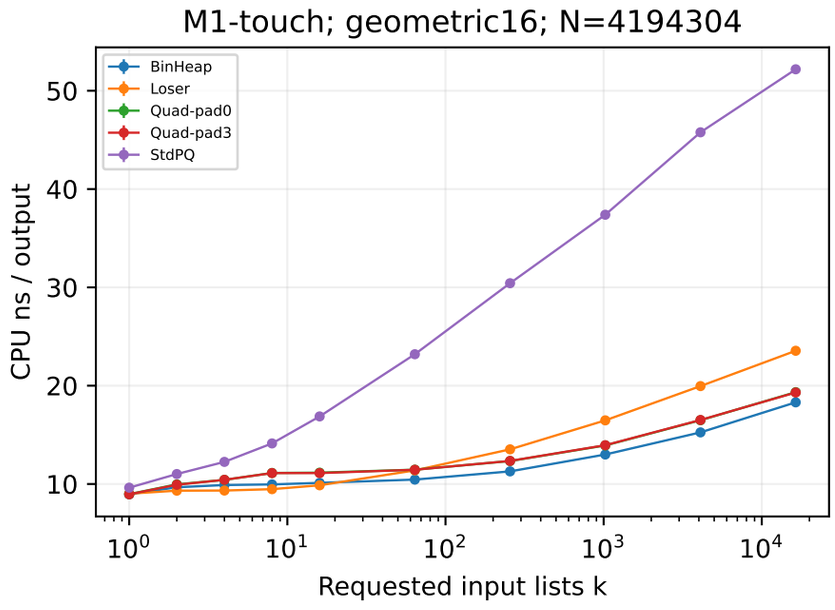
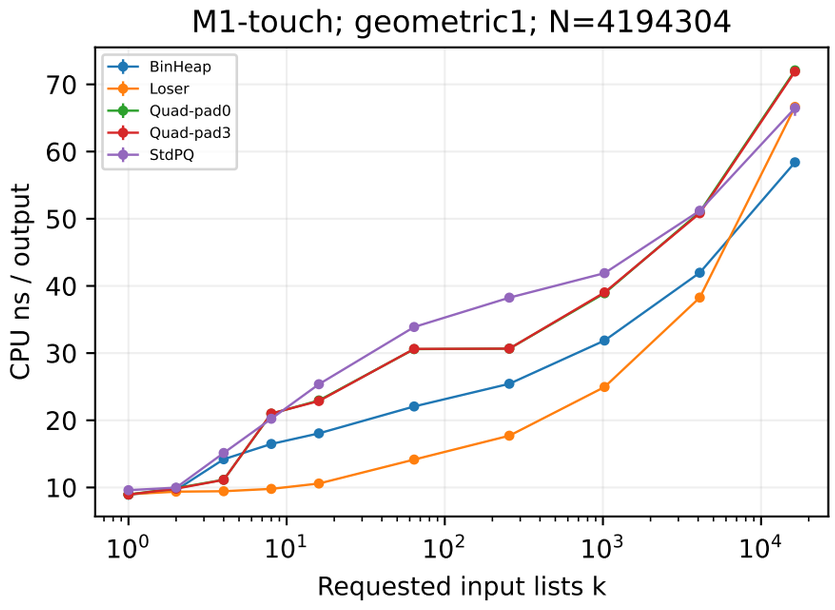
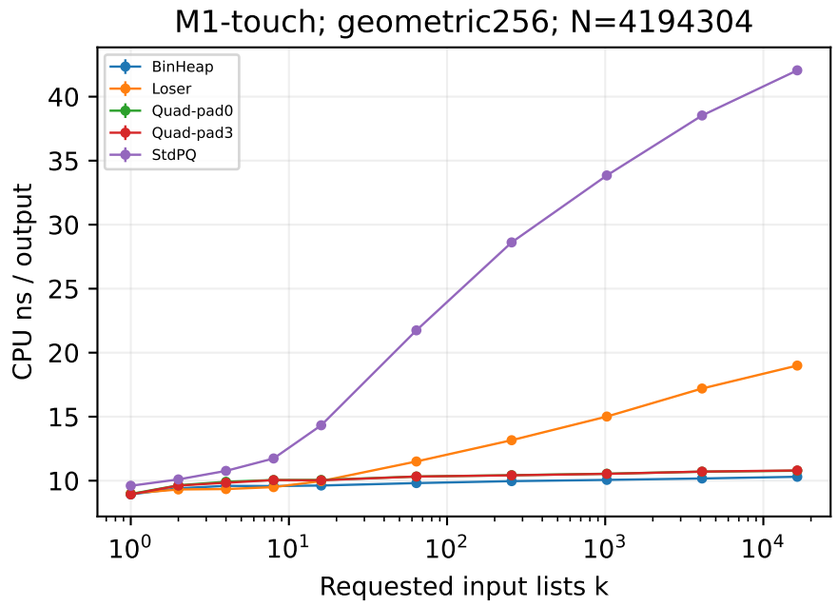
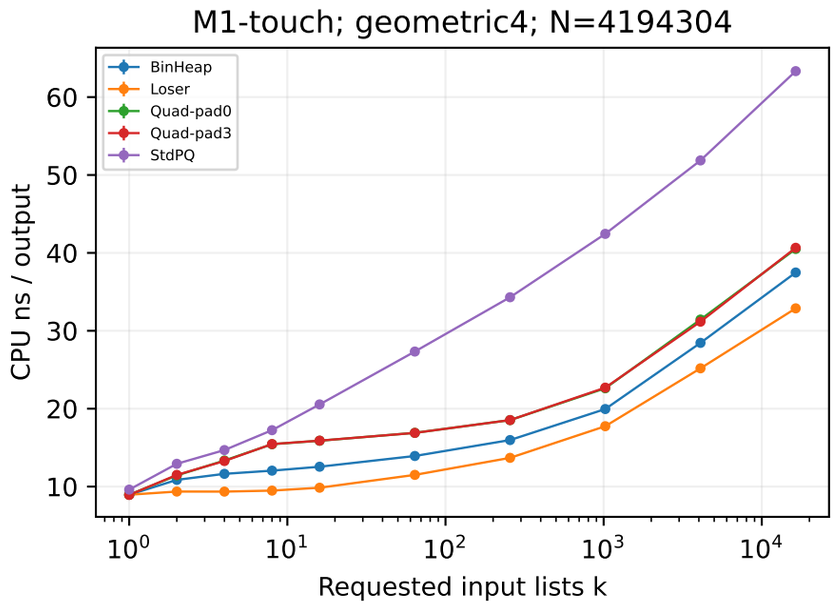
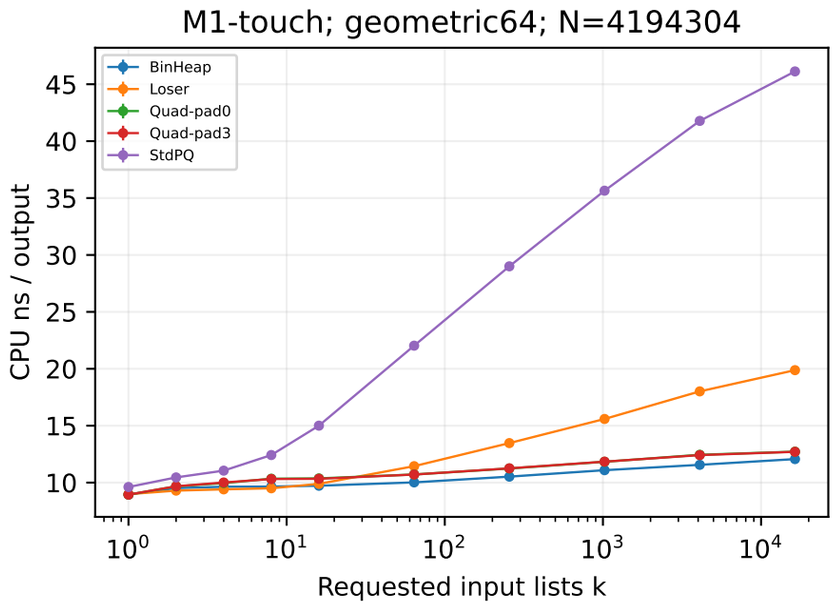
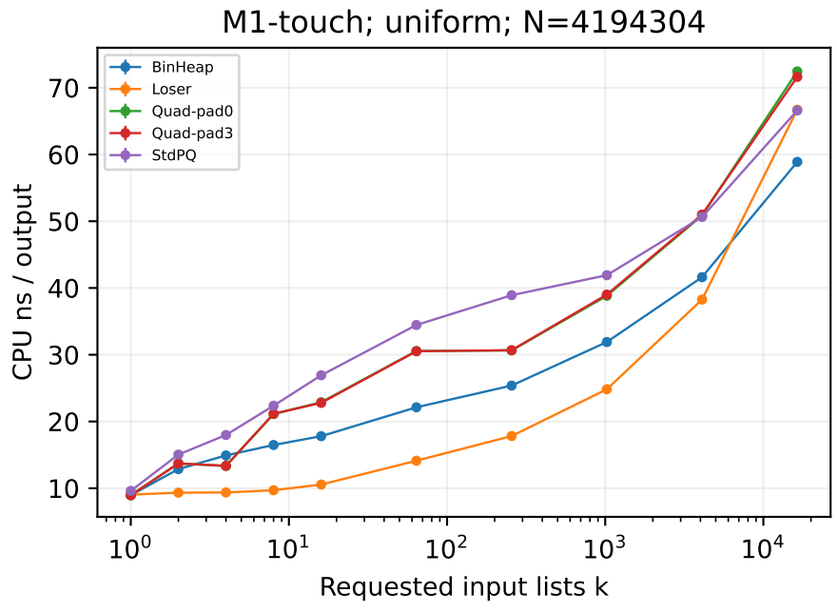


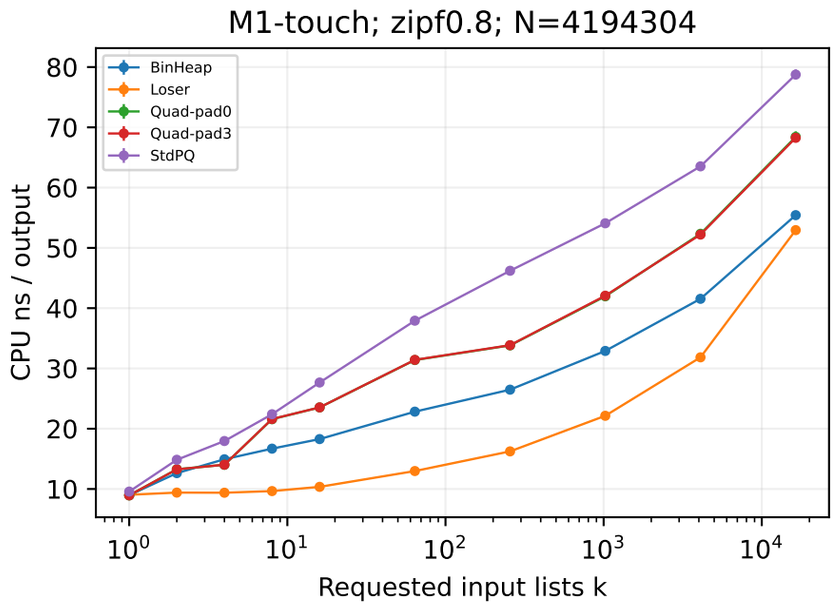


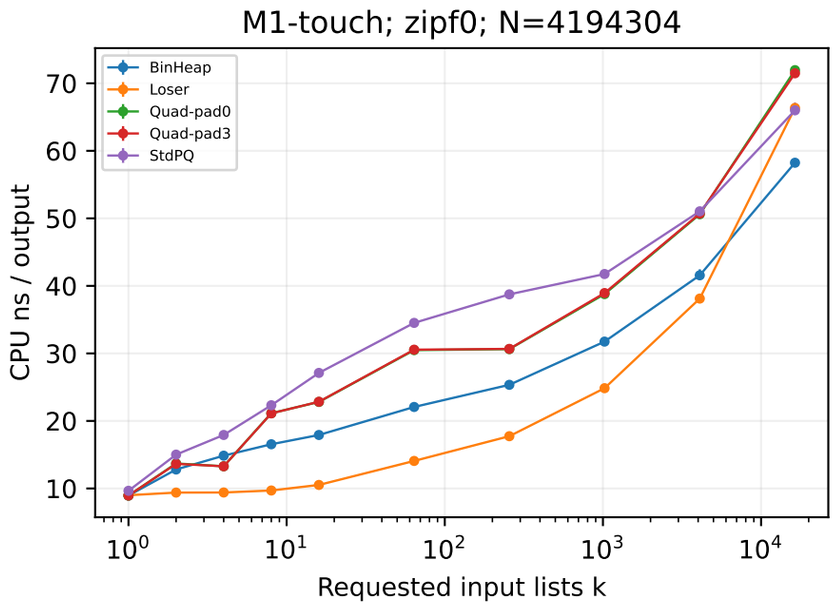


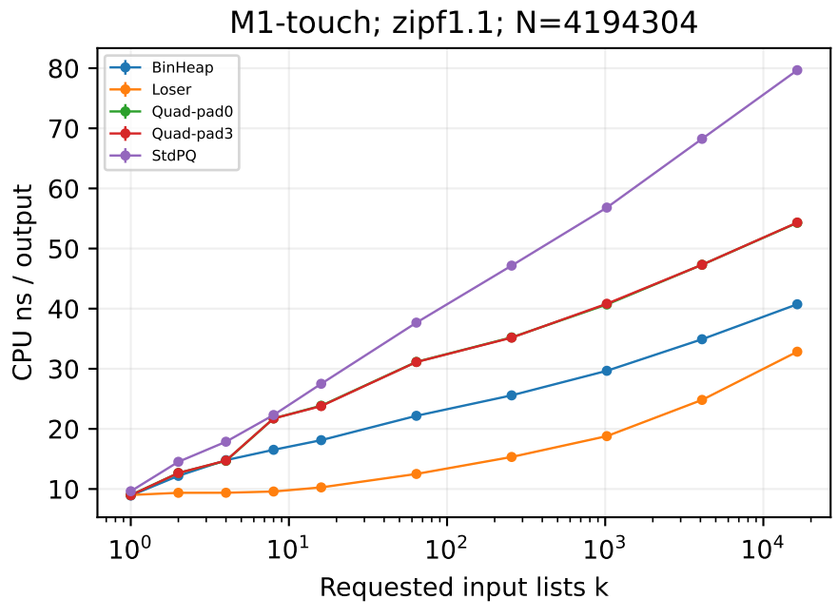


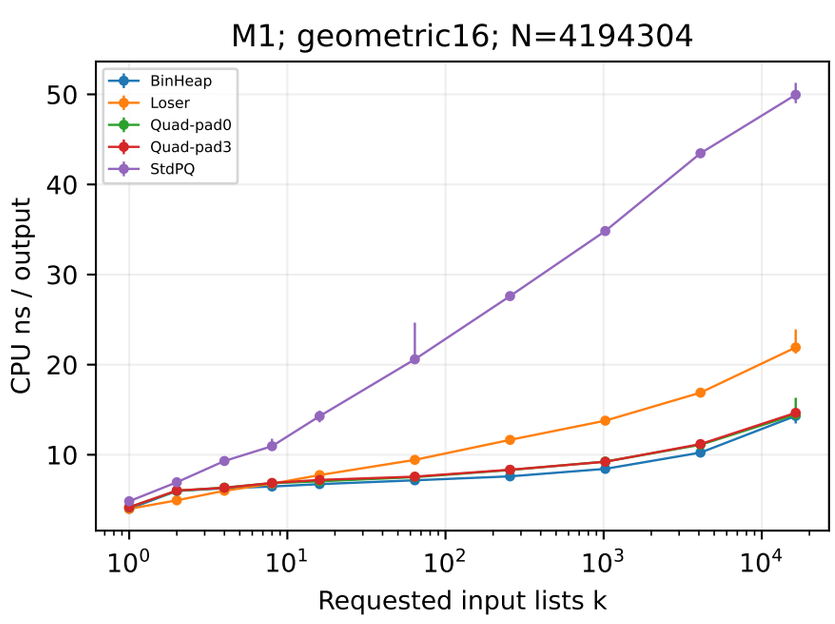


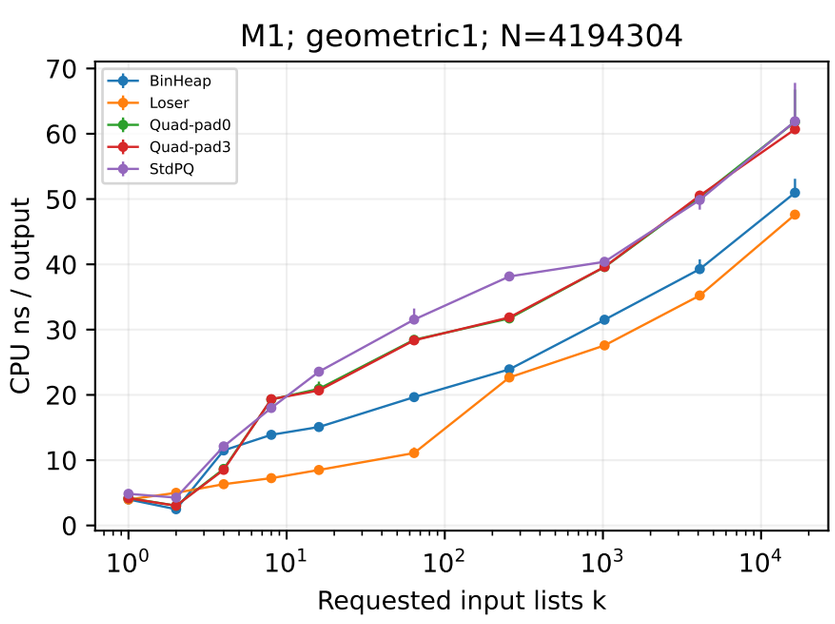


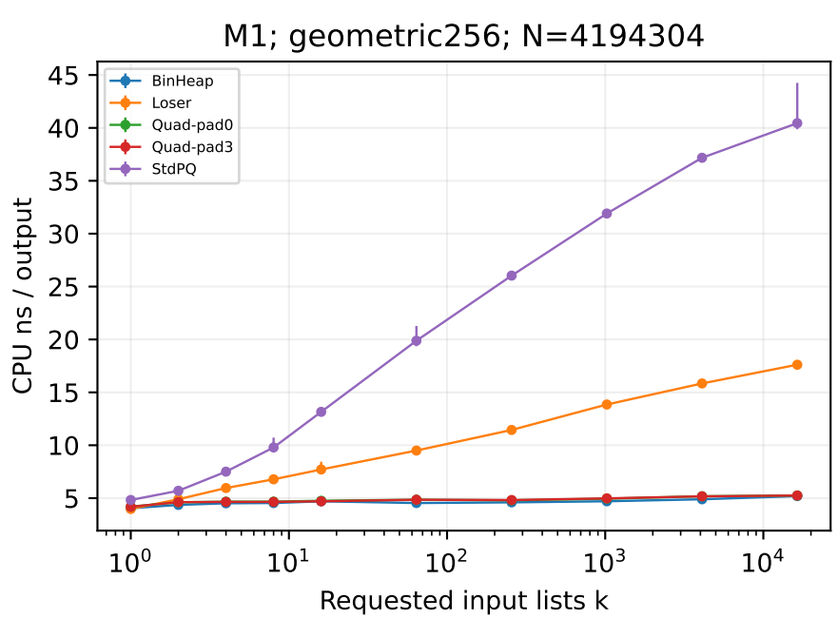


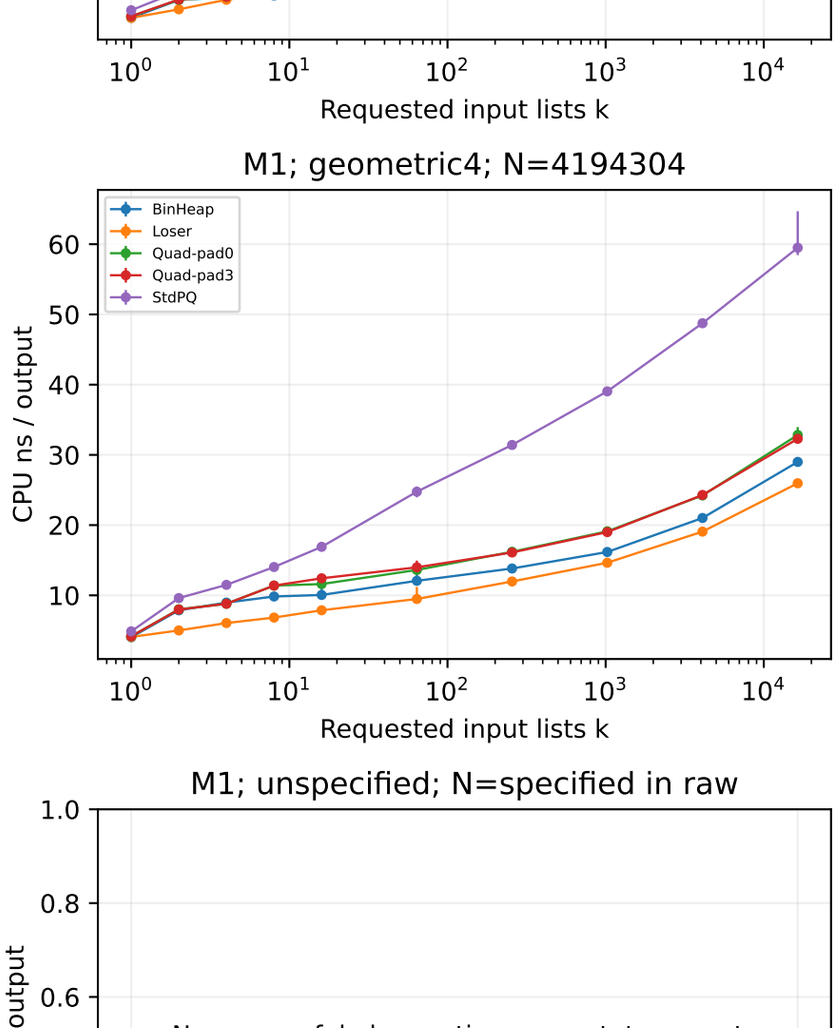


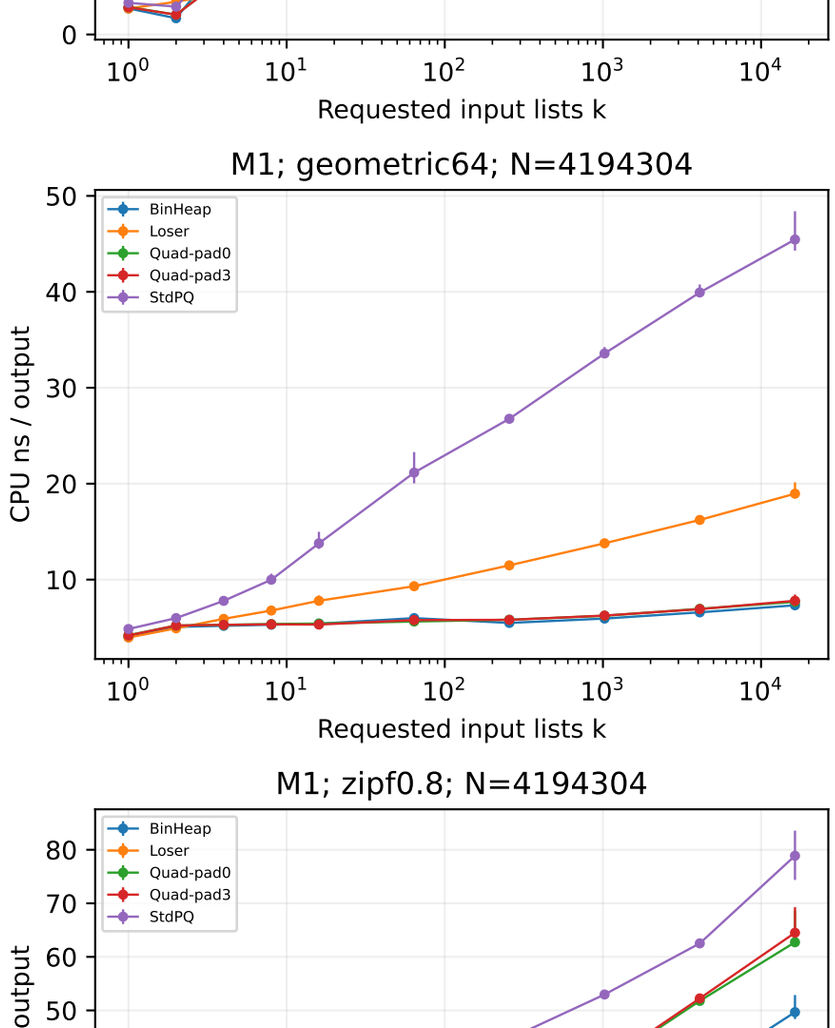


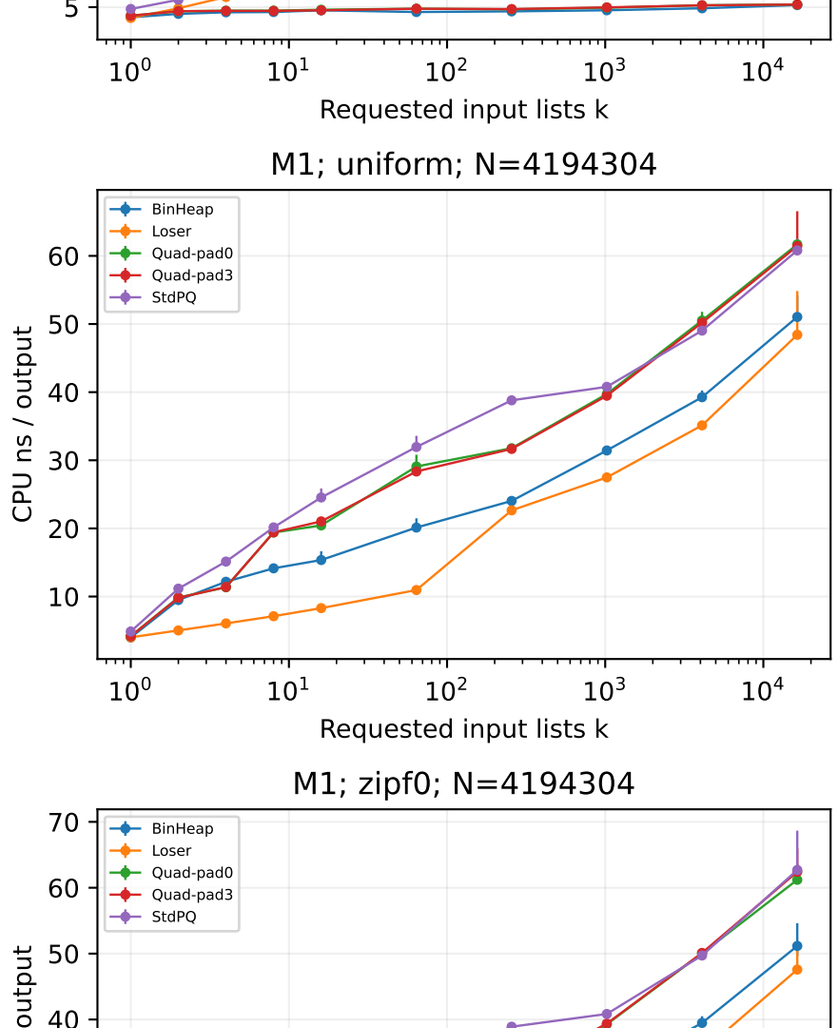


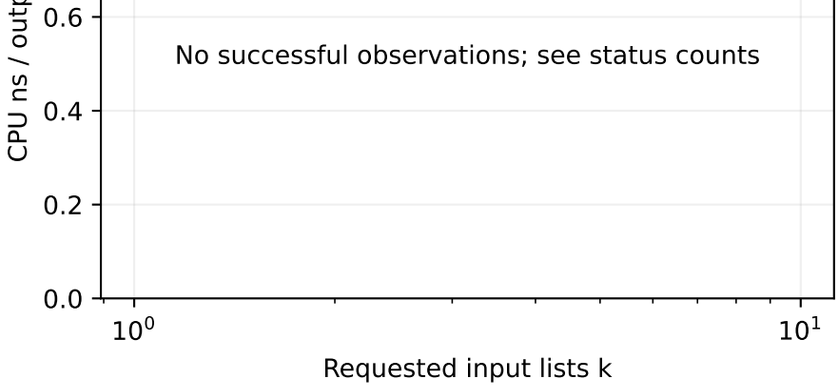


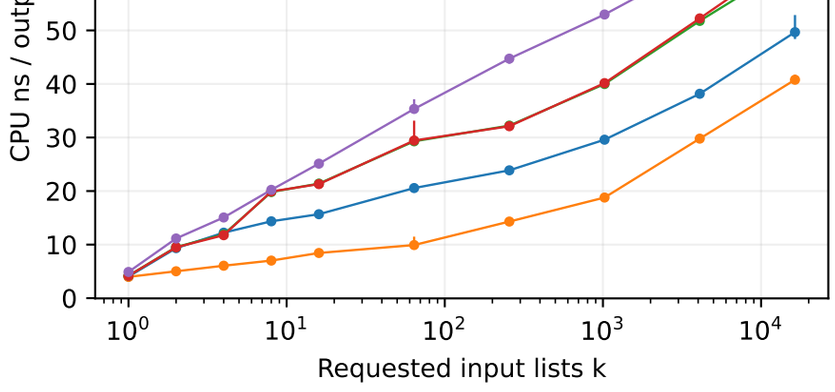


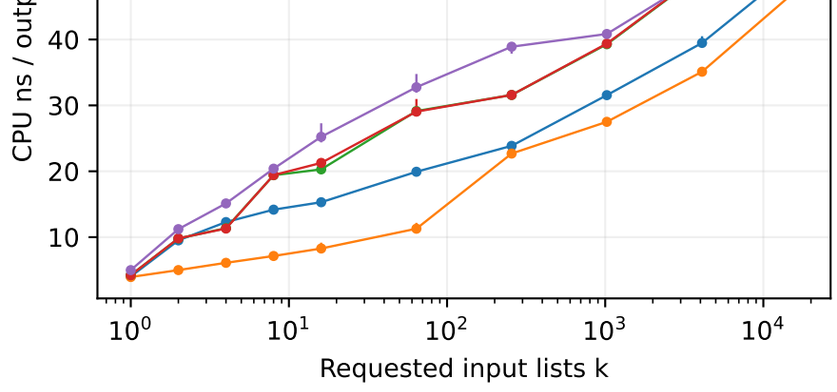


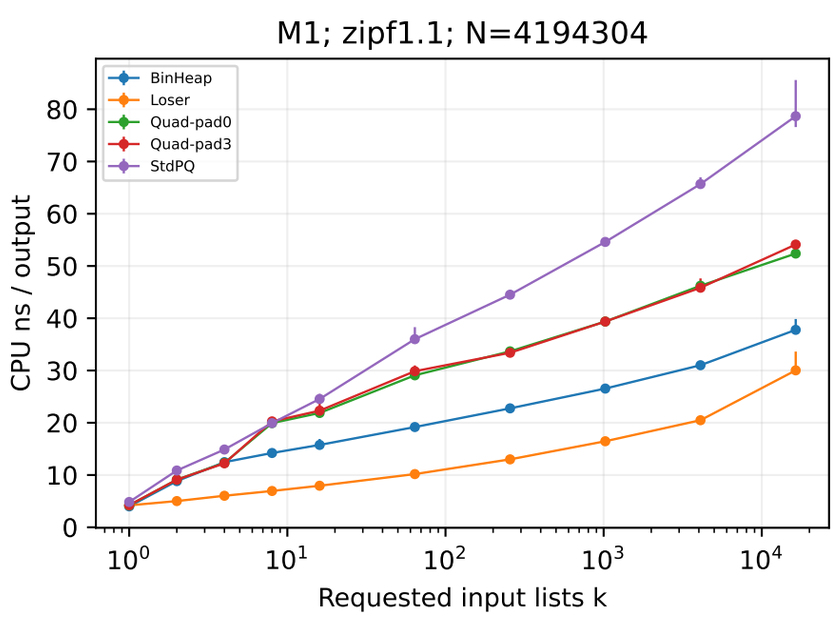












Measured outcomes: ok=13500, non-ok=1; failures retained in tip-summary.json. Error bars: seeded 10,000-resample 95% median bootstrap CIs.